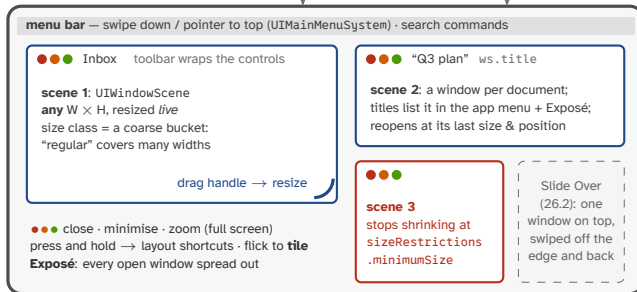


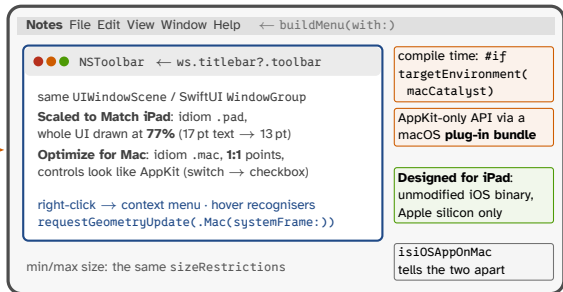
**In one line:** In iPadOS 26 a scene can be a **freely resizable window** with Mac-style **window controls**, so an iPad app must lay out for **any width and height**, not a few Split View fractions — `UIRequiresFullScreen` is **deprecated**. **Mac Catalyst** recompiles the **same** UIKit scenes into a Mac app (title bar, NSToolbar, menu bar); **Designed for iPad** runs the iOS binary as is.

## iPad — Windowed Apps

one process · UIApplication — a UIWindowSession per window



## Mac — the same scenes under Catalyst



## iPadOS 26

- Modes** (Settings › Multitasking & Gestures): Full Screen Apps · Windowed Apps · Stage Manager (stages, external display). The fixed Split View widths are gone: the system resizes your scene **continuously** and remembers the size.
- Clear the controls:**  
`preferredWindowingControlStyle(for:) → .automatic / .minimal / .unified`; bars use `layoutGuide(for: .margins(cornerAdaptation: .horizontal))` (26).
- Size limits:** `windowScene.sizeRestrictions` (`minimumSize`, `maximumSize`, `allowsFullScreen`) is nil where the system cannot resize — the write silently no-ops. SwiftUI: `windowResizability(.contentMinSize)`, `.defaultSize` (17).
- Live resize:**  
`windowScene(_:didUpdateEffectiveGeometry:)` (26) + `effectiveGeometry.isInteractivelyResizing`: defer costly work to the drag's end. SwiftUI `.onInteractiveResizeChange` (26).
- UIRequiresFullScreen** (deprecated 26): in a window the app is **scaled**, not resized; with the iOS 27 SDK it is resized *discretely* when the drag ends (TN3192). Use `sizeRestrictions` + `prefersInterfaceOrientationLocked`; `UIRequiresFullScreenIgnoredStartingWithVersion` for older OSes.
- iOS 26 SDK: no scaling or letterboxing on new screens. **TN3187:** in the release after iOS 26 an app without the UIWindow life cycle **won't launch**.
- New window:** `activateSceneSession(for:)` (17; `requestSceneSessionActivation` is deprecated). **External display:** `.windowApplication.role = interactive windows`; `.windowExternalDisplayNonInteractive = presentation screen`. `UISplitViewController` (26): draggable column widths, `.inspector.columm`.

## Example — one scene delegate, iPad and Mac

```
func scene(_ scene: UIWindow, willConnectTo s: UIWindowSession,
options: UIWindow.ConnectionOptions) {
    guard let ws = scene as? UIWindowScene else { return }
    ws.sizeRestrictions?.minimumSize = CGSize(width: 500, height: 400)
    ws.title = doc.name // window list, Expose, Mac title
    #if targetEnvironment(macCatalyst) // titlebar exists only on Mac
    ws.titlebar?.toolbar = NSToolbar(identifier: "main")
    #endif
}

func preferredWindowingControlStyle(for ws: UIWindowScene)
-> UIWindowScene.WindowingControlStyle { .unified } // 26

func windowScene(_ ws: UIWindowScene,
didUpdateEffectiveGeometry old: UIWindowScene.Geometry) {
    renderer.paused = ws.effectiveGeometry.isInteractivelyResizing
}

UIApplication.shared.activateSceneSession(for: .init( // 17
role: .windowApplication, userActivity: docActivity, options: nil))
```

## Mac Catalyst

- Tick the *Mac Catalyst* destination: the UIKit target is rebuilt for macOS (Intel + AS), each scene a Mac window.
- Mac idiom** costs work: fix hard-coded sizes and assets; `UIPageControl` and some customisations (button titles for non-`.normal` states, slider images) **throw**. Keep one control iPad-style: `preferredBehavioralStyle = .pad`.
- Chrome:** `windowScene.titlebar` (UITitlebar, Catalyst-only); `toolbar`, `toolbarStyle`, `titleVisibility`, `representedURL`.
- AppKit-only API** (status item): a macOS bundle inside the app, `Bundle(url:)?.load()`, its principal class behind an `@objc` protocol — a technique, not an API.

## Which way onto the Mac? (AS = Apple silicon)

| route                 | build                                          | Macs                     | looks · cost                |
|-----------------------|------------------------------------------------|--------------------------|-----------------------------|
| Designed for iPad     | none: iOS binary; opt out in App Store Connect | AS only                  | iPad UI (iPhone: fixed) · 0 |
| Catalyst, scaled      | recompile, same target + fix sizes, assets     | Intel + AS<br>Intel + AS | iPad at 77% · low           |
| Catalyst, Mac idiom   | macOS target, #if os(macOS)                    | all                      | Mac controls · medium       |
| SwiftUI multiplatform |                                                |                          | native · medium             |
| AppKit                | separate UI code                               | all                      | native · high               |

## Traps

- ! "Size classes are enough" — a *regular* window can be almost any width: lay out from bounds, set a minimum.
- ! Layout on `userInterfaceIdiom == .pad`: wrong in a narrow iPad window and under the Mac idiom.
- ! `UIRequiresFullScreen` as the fix: deprecated, scaled in windows, discrete with SDK 27.
- ! `isMacCatalystApp` is true for Designed-for-iPad apps too — use `isiOSAppOnMac`.
- ! `titlebar` without `#if targetEnvironment(macCatalyst)` — the iOS build fails.

## Remember

**"Any size, any number, any screen: lay out by bounds, name every window; on the Mac pick 77% or 1:1."**